

Lecture#06

Random Variable

A variable, whose values are any definite numbers or quantities that arise as result of chance factors such that they cannot exactly be predicted in advance, is called a random variable.

The random variable may also be viewed as a function.

For example, A school consists of 7 teachers of whom 4 are males and 3 are females. A committee of 2 teachers is to be formed. If Y stands for the number of male teachers selected, then Y is a random variable assuming the values of 0, 1 and 2. The possible outcomes and the values of the random variable Y are:

Events	Sequence of events	$Y = y$
e_1	Male, male	2
e_2	Male, female	1
e_3	Female, female	0

Types of random variable:

A random variable may be classified as either discrete or continuous.

Discrete random variable:

A random variable defined over a discrete sample space (i.e. that may only take on a finite or countable number of different isolated values) is referred to as a discrete random variable.

Example:

- Number of telephone calls received in a telephone booth per day.
- Number of under five children in a family.

Continuous random variable:

A random variable defined over a continuous sample space (i.e. which may take on any value in a certain interval) is referred to as a continuous random variable.

Example:

- Weight of a six-month-old baby.
- Longevity of an electric bulb.

Probability Distribution:

Any statement of a function associating each of a set of mutually exclusive and exhaustive classes or class intervals with its probability is a probability distribution. A probability distribution will be either discrete or continuous according to the random variable of interest. It is a mathematical function that describes the likelihood of obtaining the possible values that a random variable can take. It provides a systematic way to assign probabilities to different outcomes, reflecting the uncertainty associated with random phenomena. Probability distributions can be categorized into two main types: discrete and continuous.

Types of Probability Distributions:

1. **Discrete probability distribution:** A distribution where the random variable can only take distinct, separate values. Examples- Bernoulli distribution (models a binary outcome), binomial distribution (describes the number of successes in a fixed number of independent Bernoulli trials), Poisson distribution (models the number of events occurring in a fixed interval of time or space).
2. **Continuous probability distribution:** A distribution where the random variable can take any value within a given range. Examples- uniform distribution (all values within a given interval are equally likely), normal distribution (symmetric bell-shaped curve; characterized by mean and standard deviation), exponential distribution (models the time until an event occurs in a Poisson process). Understanding these types of distributions is fundamental for analyzing and interpreting data in various fields, from statistics and finance to science and engineering.

Example 1: A bag contains 10 balls of which 4 are black. If 3 balls are drawn at random without replacement, obtain the probability distribution for the number of black balls drawn.

Solution: If X denotes the number of black balls drawn, then clearly X can assume values 0, 1, 2 and 3. To obtain the probability distribution of X , we need to compute the probabilities associated with 0, 1, 2 and 3. Since 3 balls are to be chosen, the number of ways in which this choice can be made is $^{10}C_3$.

Thus,

$$f(0) = P(X = 0) = {}^4C_0 \times {}^6C_3 / {}^{10}C_3 = 20/120$$

$$f(1) = P(X = 1) = {}^4C_1 \times {}^6C_2 / {}^{10}C_3 = 60/120$$

$$f(2) = P(X = 2) = {}^4C_2 \times {}^6C_1 / {}^{10}C_3 = 36/120$$

$$f(3) = P(X = 3) = {}^4C_3 \times {}^6C_0 / {}^{10}C_3 = 4/120$$

Hence, the tabular form of the probability distribution of X will be as follows:

X	$f(x)$
0	20/120
1	60/120
2	36/120

Example 2: An unbiased coin is tossed three times. Let X be the number of runs obtained as a result of the outcomes of this experiment. Find the probability distribution of X .

(Note. A run is a block of identical consecutive outcomes, i.e., A run = a maximal sequence of either all heads or all tails.

Example:

Example 1 Outcome: $H H H$ All are the same. Only one block. So number of runs = 1	Example 2 Outcome: $H H T$ First you have $H H$ (one run of H's) Then it switches to T (a second run) Runs = 2	Example 3 Outcome: $T H H$ Starts with T (one run) Then switches to H H (second run) Runs = 2	Example 4 Outcome: $H T H$ $H \rightarrow T \rightarrow H$ Changes every time, so Runs = 3
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Solution: The probability distribution of X is given in the following table.

Outcome	Number of runs ($X = x$)	$P(X = x)$
$\overline{H H H}$	1	1/8
$\overline{H H T}$	2	1/8
$\overline{H T H}$	3	1/8
$\overline{H T T}$	2	1/8
$\overline{T H H}$	2	1/8
$\overline{T H T}$	3	1/8
$\overline{T T H}$	2	1/8
$\overline{T T T}$	1	1/8

The random variable X is seen to take on three distinct values 1, 2 and 3 with probabilities $2/8$, $4/8$, and $2/8$ respectively. The values of the random variables and their associated probabilities are summarized in a tabular form, which appear below:

Number of runs ($X = x$)	$P(X = x)$
1	$2/8$
2	$4/8$
3	$2/8$
Total	1.0

Discrete probability distribution or Probability mass function (pmf):

If a random variable “ X ” has a discrete distribution, the probability distribution of X is defined as the function f such that for any real numbers x , $f(x) = P(X = x)$.

The function $f(x)$ defined above must satisfy the following conditions to be a pmf.

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$
3. $P(X = x) = f(x)$

Verify whether the following functions are probability mass function.

- a. $f(x) = \frac{2x-1}{8}$, $x = 0,1,2,3$
- b. $f(x) = \frac{3x-6}{8}$, $x = 1,2$
- c. $f(x) = \frac{x+1}{16}$, $x = 0,1,2,3$

Solution:

- a. Summing the function over the entire range of X ,

$$\begin{aligned}\sum_{x=0}^3 f(x) &= f(0) + f(1) + f(2) + f(3) \\ &= -\frac{1}{8} + \frac{1}{8} + \frac{3}{8} + \frac{5}{8} = 1\end{aligned}$$

Here, the total probability is 1, but $(X = 0) = f(0) = -\frac{1}{8}$, which provides impossible negative value. So, $f(x)$ is not a probability mass function.

b. Here, $\sum_{x=1}^2 f(x) = \frac{9}{21} + \frac{12}{21} = 1$

and, $f(x) > 0$, So, $f(x)$ is a probability mass function.

c. Summing the function over the entire range of X,

$$\begin{aligned}\sum_{x=0}^3 f(x) &= \frac{1}{16} + \frac{2}{16} + \frac{3}{16} + \frac{4}{16} \\ &= \frac{10}{16} \neq 1\end{aligned}$$

Here, the total probability is not 1. So, $f(x)$ is not a probability mass function.

Example with pmf table

Problem 01. The probability mass function example is given below:

Question: Let X be a random variable and P (X = x) is the pmf given by:

X	0	1	2	3	4	5	6	7
P(X=x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

1. Determine the value of K

2. Find the probability (i) $P(X \leq 6)$, (ii) $P(3 < X \leq 6)$

Solution:

(1) We know that

$$\sum P(X_i) = 1.$$

$$\text{Therefore, } 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k - 1 = 0; (10k - 1)(k + 1) = 0$$

$$10k - 1 = 0 \text{ and } k + 1 = 0$$

Therefore, $k = \frac{1}{10}$ and $k = -1$.

$K = -1$ is not possible because the probability volume ranges from 0 to 1.

Hence the value of k is $\frac{1}{10}$.

(2)(i)

$$P(X \leq 6) = 1 - P(X > 6)$$

$$\begin{aligned}
&= 1 - (7k^2 + k) = 1 - \left(7\left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right) \right) = 1 - \left(\frac{17}{100}\right) \\
&= \frac{100 - 17}{100} \\
&= \frac{83}{100}
\end{aligned}$$

$$\text{Therefore, } P(X \leq 6) = \frac{83}{100}$$

$$\begin{aligned}
\text{(ii) } P(3 < X \leq 6) &= P(X = 4) + P(X = 5) + P(X = 6) \\
&= 3k^2 + k^2 + 2k^2 \\
&= \left(\frac{3}{10}\right) + \left(\frac{1}{10}\right)^2 + 2\left(\frac{1}{10}\right)^2 \\
&= 3/10 + 1/100 + 2/100 \\
&= (30 + 3)/100 \\
&= 33/100 \\
P(3 < x \leq 6) &= \frac{33}{100}
\end{aligned}$$

Example: The pmf of X is given as

$$\begin{aligned}
f(X) &= \alpha \left(\frac{3}{4}\right)^x, X = 0, 1, 2, \dots, \infty \\
&= 0 \text{ elsewhere}
\end{aligned}$$

Evaluate α and find $P(X \leq 3)$.

Solution:

Since $f(x)$ is a probability function, $\sum_{x=0}^{\infty} f(x) = 1$

Now,

$$\begin{aligned}
f(0) &= \alpha \left(\frac{3}{4}\right)^0 = \alpha \\
f(1) &= \alpha \left(\frac{3}{4}\right)^1 = \alpha \left(\frac{3}{4}\right) \\
f(2) &= \alpha \left(\frac{3}{4}\right)^2 \\
f(3) &= \alpha \left(\frac{3}{4}\right)^3
\end{aligned}$$

And so on.

Hence,

$$\begin{aligned}\sum_{x=0}^{\infty} f(x) &= \alpha + \alpha \left(\frac{3}{4}\right) + \alpha \left(\frac{3}{4}\right)^2 + \alpha \left(\frac{3}{4}\right)^3 + \dots = 1 \\ \Rightarrow \alpha \left(1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots\right) &= 1 \\ \Rightarrow \alpha \left(\frac{1}{1 - \frac{3}{4}}\right) &= 1 \\ \Rightarrow \alpha \times 4 &= 1 \\ \Rightarrow \alpha &= \frac{1}{4}\end{aligned}$$

So, the complete *pmf* of X is $f(x) = \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^x, x = 0, 1, 2, \dots, \infty$

$$\begin{aligned}\text{Now, } P(X \leq 3) &= P(X = 3) + P(X = 2) + P(X = 1) + P(X = 0) \\ &= \frac{1}{4} \left(1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3\right) \\ &= \frac{175}{256}.\end{aligned}$$

Exercises.

1. A random variable X takes the values 1, 2, 3, 4. Its pmf is

$$f(x) = kx; \quad x = 1, 2, 3, 4.$$

(i) Find k , (ii) Find $P(X = 3)$, and Find $P(X > 2)$.

2. A random variable X has pmf:

$$P(X = x) = k(x^2 + 1); \quad x = 0, 1, 2, 4.$$

(i) Find k , (ii) Find $P(X = 2)$, (iii) Find $P(X \geq 2)$ and (iv) Find $P(X \leq 1)$.

3. A random variable X has pmf:

$$f(x) = k(x + 2); \quad x = -1, 0, 1, 2, 4.$$

(i) Find k , (ii) Find $P(X < 1)$, and Find $P(X \geq 0)$.